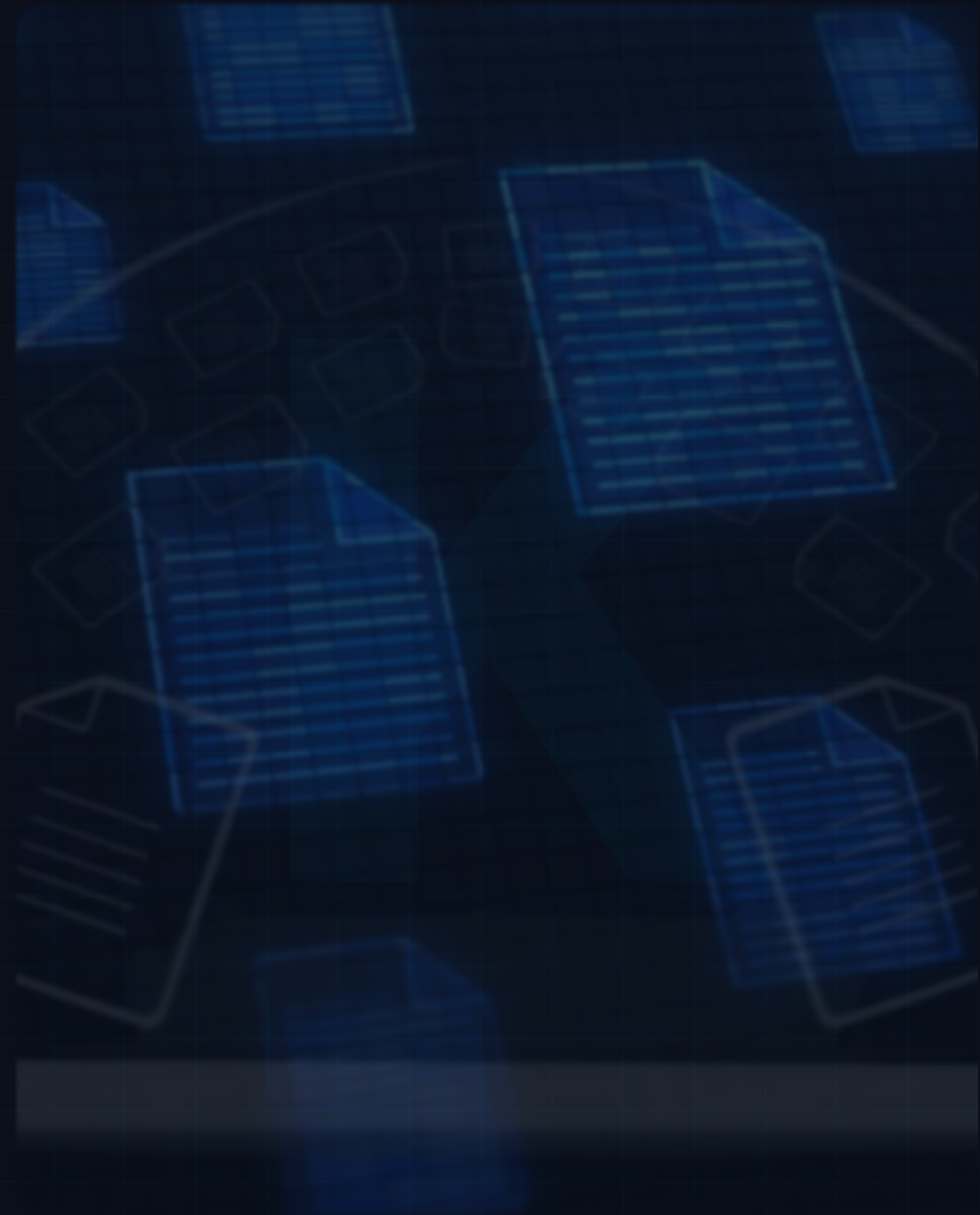


University Project Presentation

KnowledgeOS

AI Personal Knowledge Workspace

Notion + Obsidian + ChatGPT — but everything is auto-connected
by AI.



Project Overview



What It Does

KnowledgeOS is a web platform that turns scattered information into an interconnected, searchable knowledge base. Users upload documents, paste URLs, or import content — AI processes everything, builds relationships between concepts, and gives users a graph-connected knowledge library.



Purpose

To eliminate information fragmentation and give students, researchers, and professionals a unified AI-powered second brain.



Target Users

Students, researchers, professionals, content creators, and lifelong learners.

CORE PHILOSOPHY

Process → Analyze → Store



Privacy-first

No permanent file storage



Free-tier-friendly

\$5-10/month



AI-auto-connected

Knowledge

The Problem: Knowledge Fragmentation



Scattered Sources

Information lives across PDFs, web pages, notes, and documents. No single place to find or connect them.



No Semantic Understanding

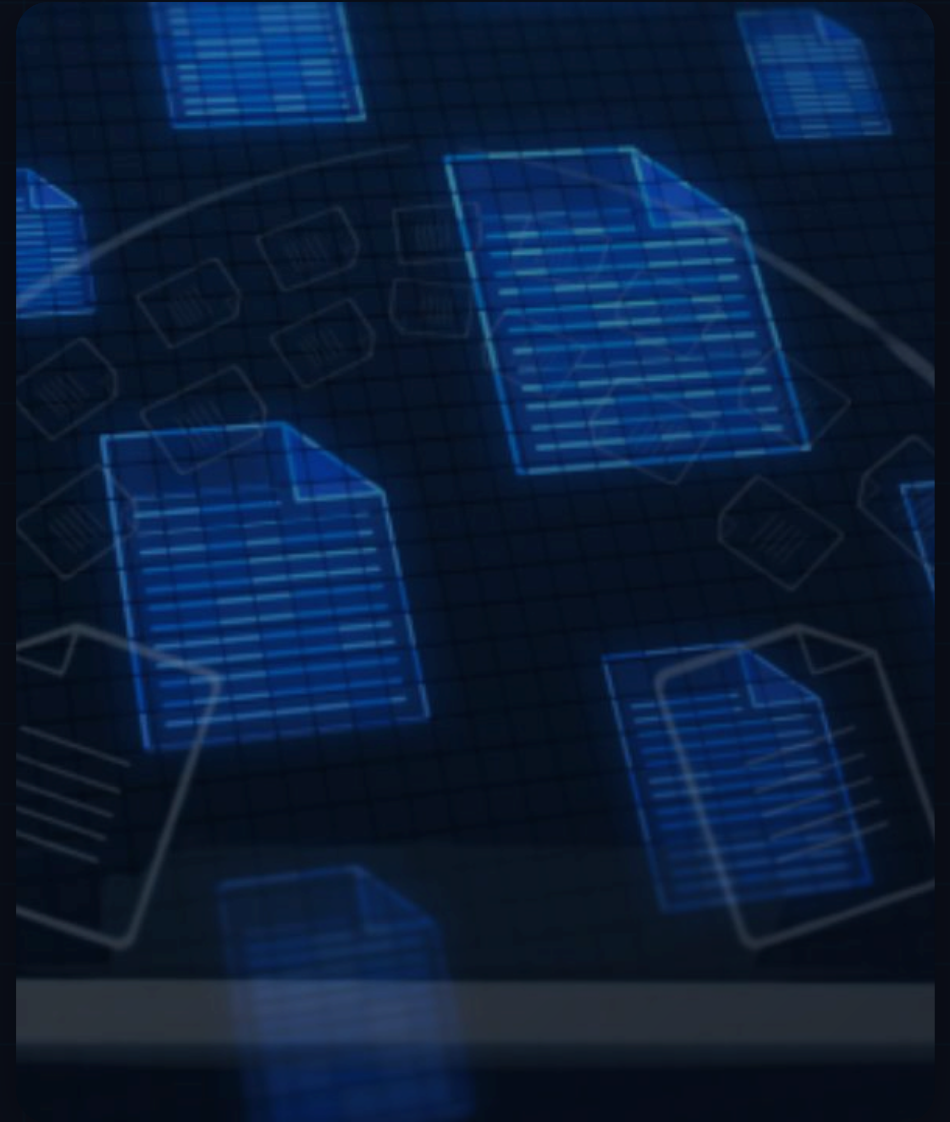
Traditional tools store files but don't understand what's inside them. Search is keyword-based, not meaning-based.



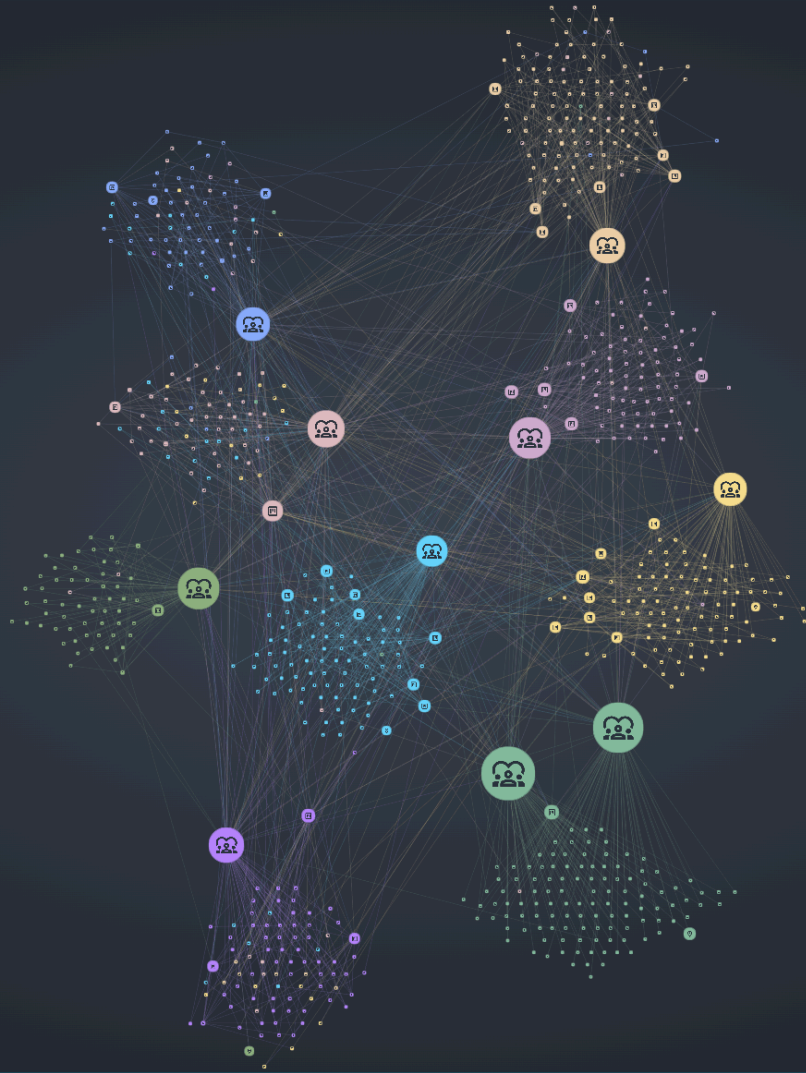
Lost Connections

Related concepts across different documents remain invisible. Users miss connections that could accelerate learning and research.

"Students and researchers waste hours re-reading material they've already covered, simply because they cannot find or connect their existing knowledge."



Our Solution: KnowledgeOS



Unified Ingestion

Upload PDFs, DOCX, PPTX, TXT, or paste any URL. One platform for all knowledge sources.



AI-Powered Understanding

Every document is automatically summarized, tagged, and mined for entities (people, topics, dates, organizations) using DeepSeek and many more.



Visual Knowledge Connections

An interactive graph reveals hidden relationships between documents and concepts — the X-factor that turns data into insight.

Unique Value: Not just storage — an AI layer that understands, connects, and lets you chat with your knowledge.

Business Requirements

User Requirements

- ✓ User registration and secure authentication (email/password)
- ✓ Upload documents in multiple formats (PDF, DOCX, TXT, MD, PPTX)
- ✓ Import web pages via URL
- ✓ Browse, search, and view processed knowledge
- ✓ Chat with their knowledge base using natural language
- ✓ Visualize knowledge connections in an interactive graph

Functional Requirements

- ✓ Text extraction from uploaded files and web pages
- ✓ AI-powered summarization (200–300 words per document)
- ✓ Entity extraction (people, topics, orgs, dates, concepts)
- ✓ Keyword/tag generation (5–10 per document)
- ✓ Knowledge graph generation with entity overlap detection

Non-Functional Requirements

- ✓ Privacy-first: no permanent file storage
- ✓ Processing time < 10 seconds per document
- ✓ Chat response time < 5 seconds
- ✓ Secure authentication via Supabase Auth
- ✓ Responsive UI with Tailwind CSS + HTMX
- ✓ Deployable on free-tier infrastructure (\$5–10/month total)

Core MVP Features

Five P0 features powering the first version



Document Upload & Processing

★★★★★ 2/5

Supports PDF, DOCX, PPTX, TXT, MD. Extracts text, generates AI summary, entities, and keywords. Files processed in memory only — never stored permanently.



Website URL Import

★★★★★ 3/5

Paste any URL. System fetches the page, strips noise (ads, nav, sidebars), extracts main content, and processes it identically to uploaded documents.



AI Processing Pipeline

★★★★★ 4/5

The brain of the system. Every content piece goes through a single optimized OpenAI call: summarization, entity extraction, and keyword generation — all in one API request.



Chat with Knowledge (RAG)

★★★★★ 4/5

A ChatGPT-style interface where AI answers questions based ONLY on the user's uploaded knowledge. Includes clickable citations to source documents and conversational memory within sessions.



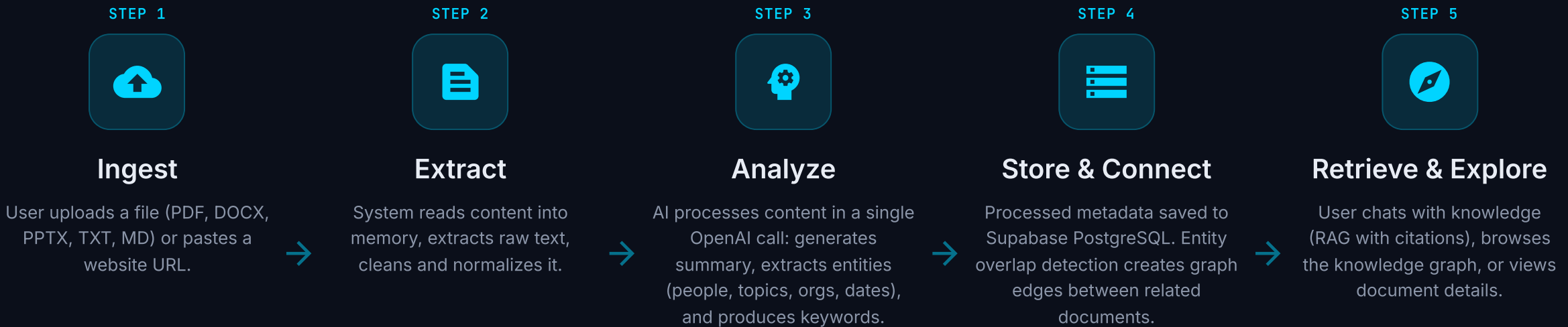
Knowledge Graph Explorer

★★★★★ 5/5

The visual wow moment. An interactive force-directed graph showing how all documents and concepts are connected. Color-coded nodes (documents, people, topics), draggable, zoomable. Reveals knowledge connections users never knew existed.

X-FACTOR FEATURE

Process Flow: End-to-End Workflow



 **Privacy Note:** Original files are processed in memory and immediately discarded. Only extracted metadata and AI-generated insights are persisted.

AI Processing & RAG Architecture

✦ AI Processing Pipeline

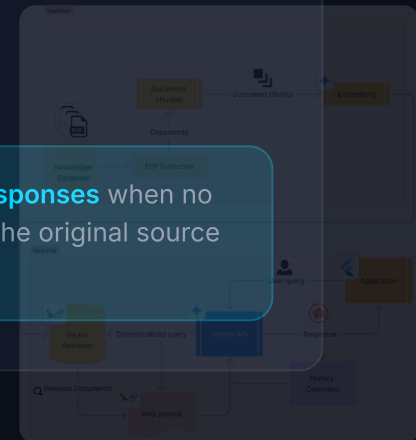
- S1** **Summarization:** Generates a concise 200–300 word summary of the content.
- S2** **Entity Extraction:** Identifies people, topics, organizations, dates, and technical terms.
- S3** **Keyword Generation:** Produces 5–10 relevant tags for searchability.

All 3 steps execute in a **SINGLE OpenAI API call** (GPT-4o-mini) using a structured JSON prompt — minimizing cost and latency.

💬 Chat with Knowledge (RAG)

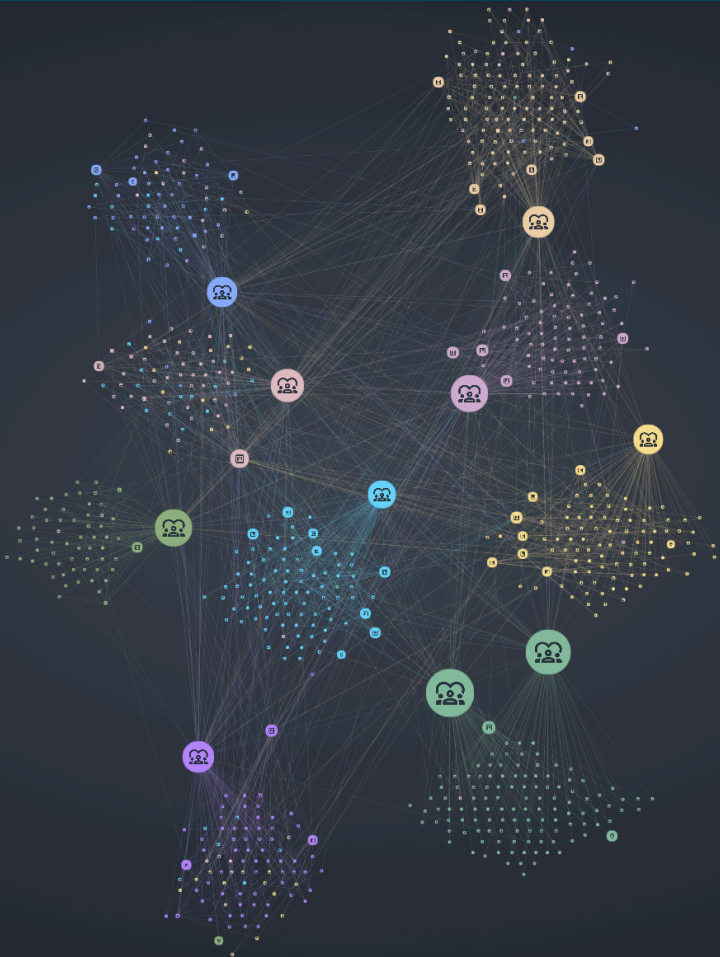
- 1** User asks a question in natural language
- 2** System finds the most relevant document chunks from the knowledge base
- 3** Chunks + question sent to GPT-4o-mini as context
- 4** AI generates answer with clickable citations to source documents

Conversational memory within sessions. **"I don't know" responses** when no relevant docs found (no hallucination). Citation links open the original source document.



Knowledge Graph Explorer

X-FACTOR FEATURE



How It Works

When a new document is added, its extracted entities are compared against all existing entities. Overlapping entities create edges (connections) between documents. Edge strength is based on how many entities two documents share.

Visual Experience

Force-directed graph layout (D3.js/vis.js). Nodes are color-coded by type — blue for documents, green for concepts, orange for people. Users can drag nodes, zoom in/out, and click any node to see its connected documents.

Why It Matters

It visualizes knowledge connections users never knew existed. It reveals hidden relationships between lecture notes, research papers, and web articles — making the platform feel like a true second brain.

Expected Outcomes & Success Metrics

For Users

- Unified knowledge repository — no more scattered information
- AI-powered insights without manual effort
- Discover hidden connections between study materials
- Chat-based Q&A with cited source references
- Privacy-first: personal data never permanently stored

For Evaluation

- Demonstrates practical AI application (RAG, NLP)
- Full-stack web development (FastAPI + Supabase + Tailwind)
- Interactive data visualization (knowledge graph)
- Cost-efficient architecture (\$5-10/month total)
- Clean, documented, testable codebase

SUCCESS METRICS

< 10s

Document Processing

< 5s

Chat Response Time

< 3s

Graph Rendering (100+ nodes)

\$5-10

Monthly Operating Cost

Thank You

KnowledgeOS — AI Personal Knowledge Workspace

Questions & Discussion

[FastAPI + Supabase](#)

[DeepSeek](#)

[Tailwind + HTMX](#)